

Aprendizagem Computacional (AC) Machine Learning (ML)



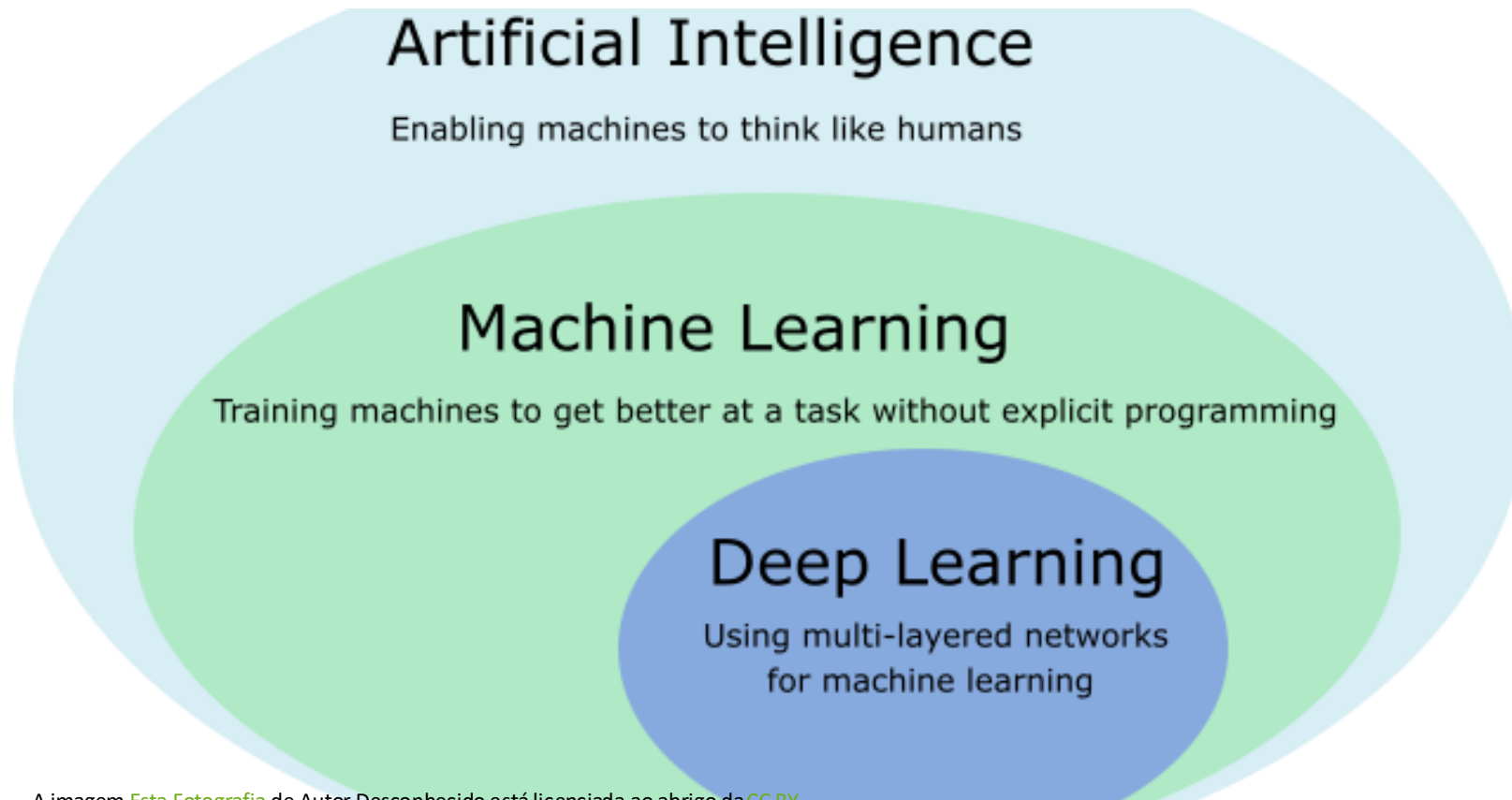
Course Presentation

César Teixeira, DEI-FCTUC, University of Coimbra
2026-2027

Machine Learning: Teaching Team

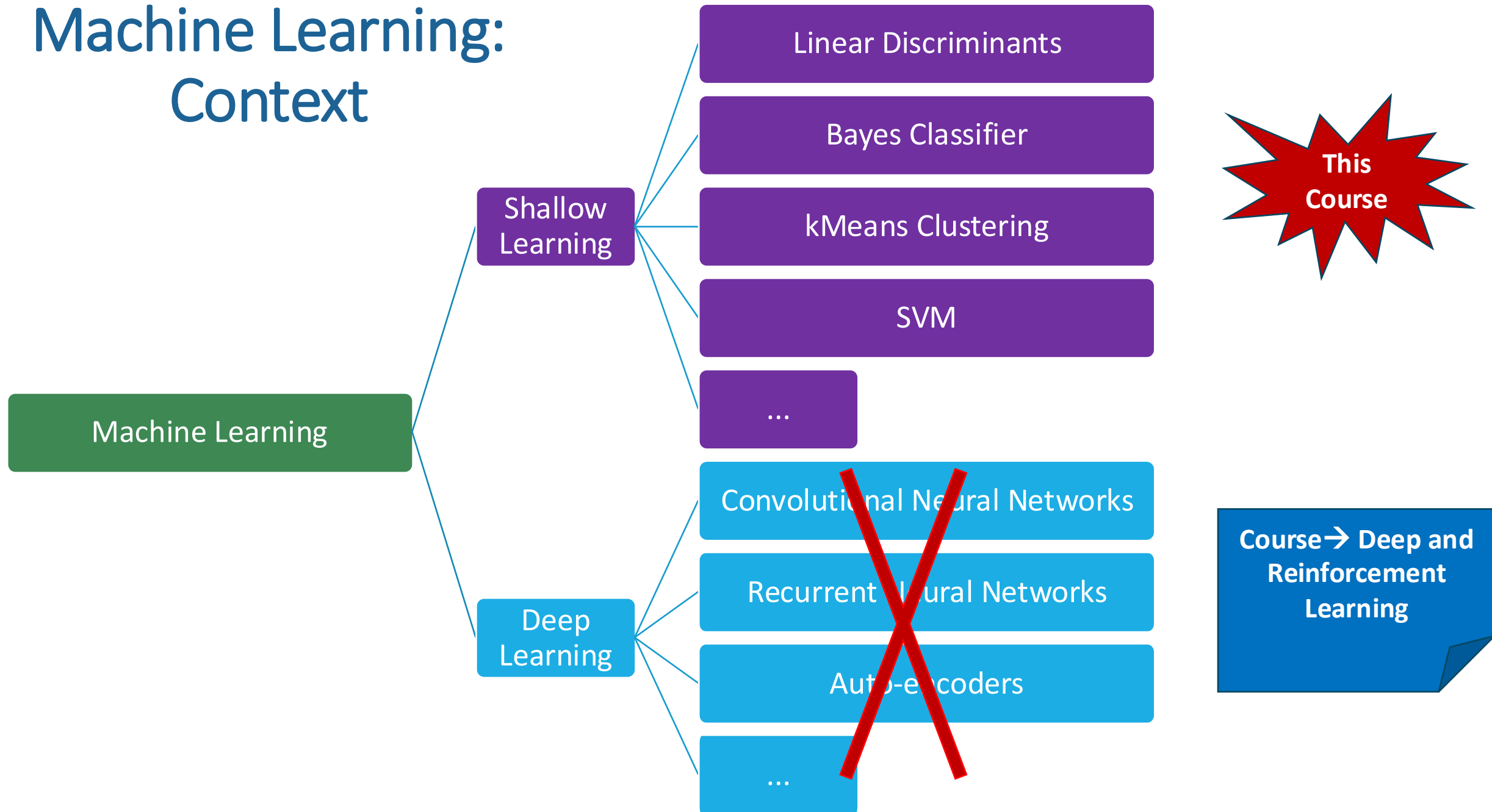
- **César Teixeira (TPs & PL1)**
 - Email: cteixei@dei.uc.pt
 - Office: **D.2.22 – DEI**
 - Office hours (**Schedule by e-mail**): **Tuesday, 10h00-12h00**
- **Gabriel Cortês (PL2)**
 - Email: cortes@dei.uc.pt
 - Office: **D2.24– DEI**
 - Office hours (**Schedule by e-mail**): **Thursday, 14h00-16h00**
- **Pedro Louro (PL3)**
 - Email: pedrolouro@dei.uc.pt
 - Office: **D2.23– DEI**
 - Office hours (**Schedule by e-mail**): **Wednesday, 17h00-18h00**

Machine Learning: Context



A imagem [Esta Fotografia](#) de Autor Desconhecido está licenciada ao abrigo da [CC-BY](#)

Machine Learning: Context



Course Outline

➤ **Part 1: Introduction.**

- i. Fundamentals & concepts.
- ii. Pipeline & phases of a ML Project.
- iii. ML spaces & transition among spaces
- iv. Generalized decision functions
- v. Hyperplane Separability
- vi. Evaluation metrics
- vii. Data partitioning

➤ **Part 2: Topics on feature engineering.**

- i. Feature Extraction
- ii. Feature Assessment & graphical Inspection
- iii. Feature Space Metrics
- iv. Feature Normalisation
- v. Curse of Dimensionality.
- vi. Dimensionality reduction
 - i. Feature selection (Kruskal Wallis & ROC curves)
 - ii. Space transformations (PCA & LDA)

Course Outline

➤ **Part 3: Supervised learning.**

- i. Linear Discriminants.
 - i. Fisher LDA
 - ii. Minimum distance classifiers
- ii. Bayesian classification
- iii. kNN
- iv. SVM
- v. Rule-based classification-Decision Trees

➤ **Part 4. Classifier combination**

- i. Bagging-Random Forests
- ii. Boosting-AdaBoost
- iii. Stacking

Course Outline

➤ **Part 5. Unsupervised learning**

- i. Partitioning clustering
 - i. kMeans
- ii. Hierarchical clustering
 - i. Divisive
 - ii. Additive
- iii. Density-based clustering
 - i. DBSCAN
 - ii. Gaussian mixed models

Course Learning Goals

✓ Knowledge and Comprehension

1. Shallow learning methods in general
2. Feature extraction and selection
3. Supervised classification
4. Unsupervised classification
5. Classifiers assessment

✓ Knowledge Application

1. Capacity of integration of acquired knowledge able to be applied in real scenarios (Medicine, Internet, Finance and Economy, Industry, ...)

✓ Judgment and Decision

1. Capacity to evaluate the most appropriate solutions for real-world problems
2. Evaluate solutions and select the best ones

✓ Communication Skills

1. Written Component (Project Report)

Course Working Model

- Total 6 ECTS
 - Theoretical-practical classes: 28 contact hours (2h/week, 1.11 ECTS)
 - Lab Classes: 28 contact hours (2h/week, 1.11 ECTS)
 - Project, Study, Office & Assessment (102 additional hours, 3.78 ECTS)
- **Presences NOT mandatory!!**

Course Assessment

- **Exams**

- Weight → 90% (18/20)
- No Consultation
- **It is forbidden to use computers, smartwatches, and cellphones!!**
- Calculator → YES
- Theoretical & Practical Questions

- **Project Report**

- Weight → 10% (2/20)
- Team → Groups of 2 students (**MAXIMUM**)



Course Assessment → Plagiarism & Fraud!



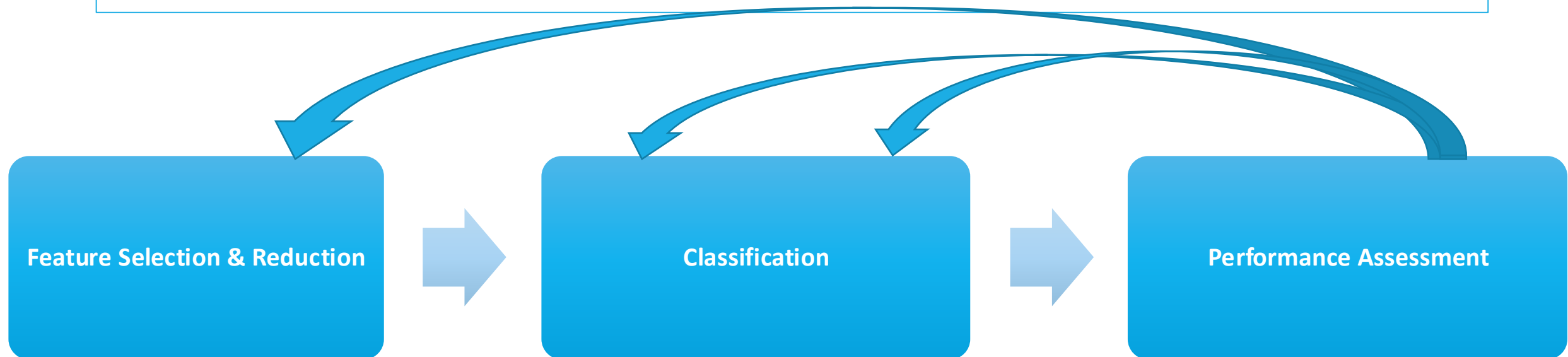
- **Read Article 126º of RAUC** (<https://diariodarepublica.pt/dr/detalhe/regulamento/945-2025-927097974>)
 - 1 - The mere possession of **items expressly prohibited by the professor** responsible for the course unit in academic exams, even if not used, results in the immediate cancellation of the assessment in which it is detected.
 - 2 - Fraud or plagiarism committed during the assessment of a course unit results in the **immediate cancellation of that assessment and leads to the immediate disqualification of the student from enrolling in the course in question**. This must be recorded on the UC's computer platform and endorsed in the student's individual file.
 - ...
- **The professor reserves the right to call any student for an additional/oral exam to clarify their written assessment.**

Course Assessment → Exams - Permitted materials!

- As an electronic device, students may only use a calculator.
- The possession of electronic devices that allow communication with the outside world, between classmates, and with the Internet is prohibited. Any attempt to use them without authorization will lead to the immediate annulment of the ongoing exam. These personal electronic devices should be enclosed inside your backpack (or other personal storing accessory)
- Only the following should be on the table:
 - the exam sheet,
 - the scratch paper,
 - the calculator, and
 - writing materials.

Project

- Real-world problem
- Machine learning design and implementation
- Platforms: Python

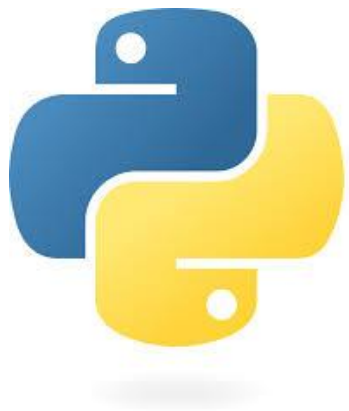


Successful Project Evaluation

- Appropriate justification for the choices made
- Proper Classifier Selection
- Effective Results Comparison & Discussion
- A Well-Written Report
- **Class participation is important!**

Evaluation Dates

- Project
 - Project Launch: **October 2nd**
 - Project Report Deadline: **December 18th**
- “Normal” Exam: **January 15th**
- “Recurso” Exam: **February 1st**



Useful Packages

- Python Numpy



- Python Scikit-Learn



- Python Pandas



- Python Plotly



- Jupyter Lab & Jupyter Notebook



Datasets

- UCI Machine Learning Datasets

<http://archive.ics.uci.edu/ml/>

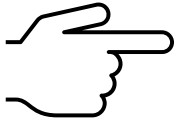
- DataSets: @UCStudent

- ...

Course Administration and Policies

- Course information, submission of assignments and project deliverables in **Inforestudante**
- Summaries, Materials, Direct Chat & Presences in **UCStudent**

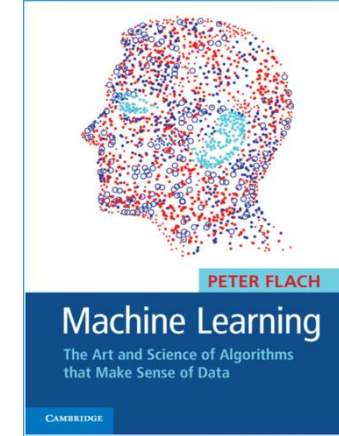
Bibliography



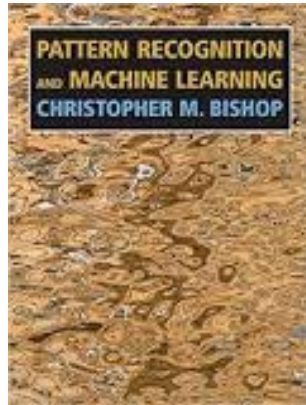
Pattern Recognition
Concepts, Methods and Applications,
Marques de Sa J.P., 2001



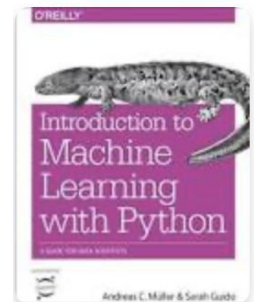
Pattern Classification,
Duda, R. O., Hart, P.E., Stork, D.,
Wiley Interscience, 2001



Machine Learning: The Art and Science
of Algorithms that Make Sense of
Data. Flach, Peter. Cambridge
University Press, 2012.



Pattern Recognition and Machine
Learning, Bishop, C.M., 2006,
Springer Verlag



Introduction to Machine Learning with
Python: A Guide for Data Scientists
Müller, Andreas
C., and Guido, Sarah. O'Reilly Media,
Incorporated, 2018.



Reconhecimento de Padrões:
Métodos Estatísticos e Neurais,
Jorge Salvador Marques 2005